SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I S.Chandra Siddhardha(192525222) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

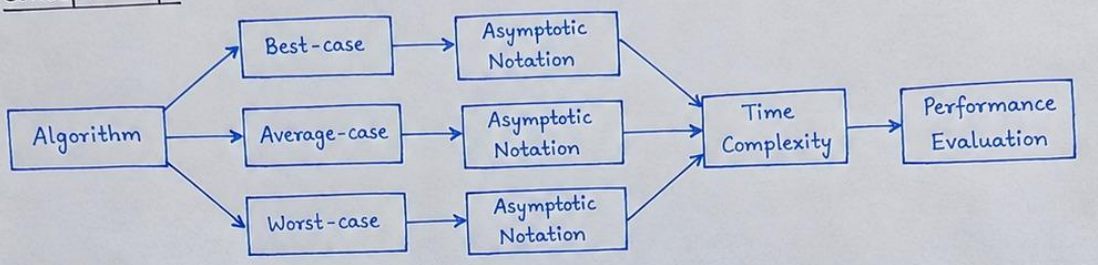
S.Chandra Siddhardha

RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

2. Consider the concept map: Algorithm → Best-case, Average-case, Worst-case → Time Complexity → Performance Evaluation. Map the relationships among the three cases and explain how they provide different views of algorithm behavior. Extend the map by linking each case to Asymptotic Notation so that the complexity can be expressed formally. Justify why all three cases are important in performance analysis rather than relying on only one measure. Interpret how the overall structure helps compare the practical and theoretical behavior of algorithms.

Concept Map:



Explanation:

1. Algorithm → Best-case, Average-case, Worst-case:
   - An algorithm can behave differently depending on the input. The best-case gives the minimum time, average-case gives the expected time, and worst-case gives the maximum time.
2. Each case → Asymptotic Notation:
   - The time taken in each case is expressed using asymptotic notations (Big-O, Big-Theta, Big-Omega) to show the growth rate formally.
3. Asymptotic Notation → Time Complexity:
   - The notation helps to express the time complexity of each case in a standard and comparable form.
4. Time Complexity → Performance Evaluation:
   - The overall time complexity, considering all cases, is used to evaluate the algorithm's performance in terms of speed and efficiency.

Justification:

- All three cases are important because an algorithm may not always perform in the average or worst case. For example, a search algorithm might be very fast for a particular input (best case) but very slow for another (worst case).
- Relying on only one measure (like best case) can be misleading and may give an unrealistic view of the algorithm's behavior.

Interpretation:

The concept map helps to see the complete picture of an algorithm's behavior. By linking each case to asymptotic notation and then to time complexity, it becomes easier to compare the practical (real-world) and theoretical (worst-case) behavior of algorithms. This structured approach ensures a reliable performance evaluation and better decision making when choosing an algorithm for a problem.